Sidhartha Reddy Potu

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Experience

Granica AI — Mountain View, CA
SWE Intern — *Sep 2025 – Mar 2026*

- Built a dual-lane FIFO load balancer for a **400+** node K8s cluster, doubling throughput (**10** to **20+** req/worker).
- Eliminated runaway K8s API calls at **20k** concurrency by caching scaler state in Go.
- Resolved lock contention and races in load-balancer goroutines, improving CPU utilization from **87%** to **97%**.
- Designed local scale-validation and integration test suites for the load-balancer and worker routing logic.

CILVR, NYU — New York, NY
Graduate Researcher, 3D Computer Vision under Prof. David Fouhey — *Jan 2024 – May 2025*

- Combined HaMeR transformer poses with COLMAP SfM into 3D hand trajectories on Epic-Kitchens/EgoExo4D.
- Optimized hand-motion trajectories via temporal smoothing with velocity and jerk regularization.
- Recovered full-perspective 3D from weak-perspective predictions using SfM intrinsics and frame transforms.
- Fine-tuned DINOv2 ViT-L/B and EfficientNet-V2 for hand–object contact classification on Hands23.
- Evaluated Depth Anything and Metric3D as auxiliary depth signals for the contact task.
- Developed visualization workflows with Rerun Viewer; scaled experiments on NYU HPC via SLURM.

Research Project, NYU — New York, NY
Egocentric Point Tracking, Embodied Learning and Vision under Prof. Mengye Ren — *Feb 2025 – May 2025*

- Evaluated DINOv2 and CroCoV2 backbones zero-shot for egocentric point tracking on EgoPoints.
- Fine-tuned both backbones self-supervised, augmenting with a delta network for keypoint displacement.

NYU — New York, NY
Course Assistant – Computer Vision, Deep Learning, LLVM — *Sep 2024 – May 2025*

- Mentored **450+** students through office hours, grading, and project advising across 3 courses.
- Assisted DL (Prof. Chinmay Hegde), LLVM (Prof. Saining Xie), and Advanced CV (Prof. David Fouhey).

Hackathons

Kube SRE Gym [GitHub] [Gallery] — Mar 2026

- Won **1st place (100+ teams)** at the OpenEnv Hackathon (PyTorch, Cerebral Valley; **$100K+** prize pool).
- Engineered an RL env where a small language model diagnoses and fixes live K8s incidents on GKE via kubectl.
- Deployed adversarial incident designer with curriculum control and LLM judge scoring SRE workflow quality.
- Post-trained Qwen-series models with GRPO via TRL on H100, using OpenEnv and HF Spaces.

Projects

Fine-Tuning Medical QA Models [GitHub] — Oct 2024 – Dec 2024

- Fine-tuned LLaMA 3.2 3B for medical QA via LoRA, QLoRA, and GaLore.
- Achieved best USMLE accuracies: Step 1 – 42% (GaLore), Step 2 – 38% (QLoRA), Step 3 – 45% (GaLore).
- Boosted MedQuAD F1 to **70%** on open-domain medical QA evaluation.

VioletPass: Ticket Booking Platform [GitHub] — Oct 2024 – Dec 2024

- Architected a scalable ticket-booking platform on AWS with Redis distributed locks to prevent double booking.
- Configured Redis TTL-based seat reservations with PostgreSQL finalization for transactional integrity.
- Implemented QR-code verification for secure and efficient event check-ins.

Education

New York University, Tandon School of Engineering — New York, NY
Master of Science in Computer Science; GPA: 3.83/4 — *Sep 2023 – May 2025*

Indian Institute of Technology (IIT), Indore — Indore, IN
Bachelor of Technology in Electrical Engineering, Minor in Astronomy; GPA: 8.49/10 — *Jul 2019 – May 2023*

Technical Skills

Languages: Python, Go, C/C++, SQL
ML & Data: PyTorch, Hugging Face (Transformers, TRL, PEFT), vLLM, TensorFlow, OpenCV, Rerun, NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, PySpark, W&B
Infrastructure: Linux, Git, Docker, Kubernetes, SLURM, Redis, PostgreSQL, AWS, GCP, CI/CD